

KEMIONSAARI VANO, Finland

WMO: 027430

Lat: 59.869N

Lon: 22.193E

Elev: 26

StdP: 14.68

Time Zone: 2.00 (EUE)

Period: 09-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	4.3	8.9	0.7	5.7	4.6	5.3	7.2	10.4	34.5	30.4	32.0	32.1	11.9	70	0.753

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.5	75.9	68.3	72.6	66.3	70.0	63.8	69.8	74.7	67.4	71.9	65.1	69.2	10.0	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
68.0	102.9	73.0	65.5	94.4	70.5	63.3	87.0	67.8	33.8	74.4	31.9	72.0	30.1	69.4	73.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 33.1	29.5	26.4	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			2.7	70.2	8.1	3.4	-3.1	72.7	-7.9	74.7	-12.4	76.6	-18.3	79.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	44.8	27.9	27.6	31.2	38.6	49.2	57.2	64.3	63.4	56.5	46.3	40.2	34.1		
	DBStd	14.20	7.80	9.29	6.38	4.55	6.47	4.36	4.97	3.50	4.28	5.21	6.36	7.41		
	HDD50	3267	684	628	583	343	96	2	0	0	7	135	296	493		
	HDD65	7448	1149	1048	1048	792	488	237	75	72	256	580	745	958		
	CDD50	1372	0	0	0	1	73	218	444	415	201	20	0	0		
	CDD65	79	0	0	0	0	0	3	54	22	0	0	0	0		
	CDH74	191	0	0	0	0	0	3	168	20	0	0	0	0		
	CDH80	15	0	0	0	0	0	0	14	1	0	0	0	0		
(14) Wind	WSAvg	13.8	15.3	14.3	14.0	12.3	11.7	12.1	11.1	11.8	13.3	15.1	16.6	17.7		
(15) Precipitation	PrecAvg	25.9	2.3	1.7	1.5	1.3	1.2	1.8	2.4	2.9	2.4	2.9	3.1	2.5		
	PrecMax	31.3	4.4	3.7	3.0	2.2	2.4	3.3	5.8	6.0	5.8	5.5	6.2	4.2		
	PrecMin	19.4	0.4	0.1	0.2	0.3	0.4	0.5	0.1	0.4	0.7	1.4	0.2	0.8		
	PrecStd	3.0	1.0	0.9	0.7	0.6	0.6	0.7	1.5	1.4	1.3	1.1	1.5	1.0		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	41.4	40.0	43.8	54.8	69.9	73.1	81.8	76.3	66.8	57.6	51.0	47.3		
		MCWB	40.0	37.2	39.1	46.0	57.7	62.8	69.7	69.3	61.9	54.0	49.3	45.2		
	2%	DB	40.1	38.5	41.4	51.0	66.1	69.4	78.5	72.4	64.7	55.8	49.8	45.3		
		MCWB	38.8	36.9	38.0	44.3	56.3	60.6	69.2	66.3	60.1	53.3	48.7	43.5		
	5%	DB	38.7	37.6	39.8	47.6	62.9	66.4	75.2	70.2	63.3	54.4	48.5	43.8		
		MCWB	37.4	36.4	37.0	42.2	55.1	59.1	68.0	64.2	59.3	52.1	47.0	42.0		
	10%	DB	37.2	36.4	38.3	44.7	59.4	64.0	72.4	68.3	61.9	53.0	47.0	42.4		
		MCWB	35.9	35.4	36.2	40.6	53.0	58.0	66.6	62.9	58.6	50.8	45.3	40.5		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	41.0	38.4	40.2	46.9	59.6	65.0	72.6	71.1	63.1	55.5	50.2	45.9		
		MCDB	41.2	39.1	42.5	53.2	66.2	71.2	77.7	75.3	65.6	56.3	50.4	46.8		
	2%	WB	39.1	37.4	38.6	44.8	57.5	62.3	70.8	68.2	61.4	54.0	48.9	43.9		
		MCDB	39.8	38.1	40.6	50.1	64.0	67.2	76.0	71.7	63.5	55.1	49.5	44.8		
	5%	WB	37.8	36.6	37.6	43.0	55.8	60.5	69.2	65.7	60.3	52.9	47.4	42.3		
		MCDB	38.6	37.4	39.2	46.9	62.1	65.0	74.3	68.6	62.4	54.2	48.4	43.6		
	10%	WB	36.1	35.6	36.6	41.3	53.8	58.9	67.1	64.3	59.3	51.2	45.7	40.8		
		MCDB	37.1	36.5	38.0	44.0	59.0	63.1	72.0	67.5	61.4	52.6	46.9	42.3		
(35) Mean Daily Temperature Range	5% DB	MDBR	6.7	6.3	7.6	8.3	9.7	8.8	8.5	7.3	6.3	5.4	4.7	5.4		
		MCDBR	5.1	4.8	6.9	12.5	14.9	12.9	12.0	9.3	7.5	5.4	4.7	5.2		
	5% WB	MCWBR	5.4	4.4	5.0	7.5	7.8	6.6	5.7	5.1	5.2	5.0	5.0	5.5		
		MCWBR	5.4	4.6	6.2	11.4	13.9	11.0	10.4	7.3	6.5	5.2	4.8	5.0		
(40) Clear-Sky Solar Irradiance	taub		0.284	0.317	0.335	0.373	0.368	0.362	0.391	0.383	0.353	0.337	0.312	N/A		
	taud		2.178	2.246	2.308	2.322	2.422	2.479	2.424	2.435	2.500	2.402	2.215	N/A		
	Ebn at Noon		154	215	252	262	273	276	264	257	247	208	140	N/A		
	Edh at Noon		14	22	28	33	33	31	32	30	24	19	13	N/A		
(44) All-Sky Solar Radiation	RadAvg		77	249	736	1269	1765	1886	1773	1393	852	367	101	49		
	RadStd		11	45	86	131	167	122	151	131	102	52	20	6		

Historical Trends

		Heating			Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
(46) Station Only	DBAvg	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47) Regional (56 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		+0.74	N/A	N/A	N/A	N/A	-239	N/A	N/A	N/A	N/A

Nomenclature: See separate page